

Dr. Jesper Armouti-Hansen

DATA SCIENTIST · ECONOMICS PHD

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Data scientist with an economics PhD. I build and evaluate statistical and machine-learning models on structured data — econometrics, model validation, calibration, and uncertainty quantification — and turn results into decisions stakeholders act on. Currently at AXA.

EXPERIENCE

Data Scientist, AXA Konzern AG
Cologne, Germany

Apr. 2026 – present

- Develop the production document-understanding pipeline that splits stacks of scanned pages into documents (boundary detection) and classifies each — hundreds of thousands of stacks per month — routing low-confidence cases to human review.
- Built document-class-specific confidence thresholds that automate more documents while holding at least 95% precision per class, cutting manual review by around 30% in production.
- Researching pipeline upgrades: an LLM post-processing step for frequently-confused class pairs, and a single vision-language model (Qwen3.5-VL 4B) to replace the two-stage BERT+CLIP and RNN stack.

Postdoctoral Researcher in Economics, University of Bonn
Bonn, Germany

Jan. 2022 – Mar. 2026

- Ran empirical and computational economics research — statistical and econometric modeling, machine-learning benchmarking, and partial-identification with simulation.
- Developed theoretical models and built reproducible analysis pipelines in Python and R; benchmarked economic theories against machine-learning models to quantify how much predictable variation each captures.
- Published peer-reviewed research and presented at international conferences.

Research Assistant in Economics, University of Cologne
Cologne, Germany

Dec. 2015 – Dec. 2021

- Conducted PhD research in microeconomic theory and experimental economics — decision-theory and contract-theory modeling, with empirical analysis of experimental data.

Intern, AirPlus International
Neu Isenburg, Germany

Dec. 2014 – May 2015

- Built VBA tools for data management and process automation.

PUBLICATIONS

80 citations · Google Scholar

1. **Efficiency Wages with Motivated Agents.** J. Armouti-Hansen, L. Cassar, A. Dereky, F. Engl. *Games and Economic Behavior*, 145, 66–83 (2024).
2. **Managing Anticipation and Reference-Dependent Choice.** J. Armouti-Hansen, C. Kops. *Journal of Mathematical Economics*, 112, 102988 (2024).
3. **Optimal Contracting with Endogenous Project Mission.** J. Armouti-Hansen, L. Cassar. *Journal of the European Economic Association*, 18(5), 2647–2676 (2020).
4. **This or That? Sequential Rationalization of Indecisive Choice Behavior.** J. Armouti-Hansen, C. Kops. *Theory and Decision*, 84(4), 507–524 (2018).

EDUCATION

PhD in Economics (Dr. rer. pol.)
University of Cologne — *Summa cum laude*

2015 – 2021

MSc International Economics & Public Policy
University of Mainz — GPA 1.6

2012 – 2014

BA Financial Management & Services
Copenhagen Business Academy

2008 – 2012

SELECTED PROJECTS

CitiBike Demand, Risk & Net Flow

Operational analysis

A per-trip crash-risk measure for a bike-share network — NYPD collisions over trip exposure, empirical-Bayes smoothed — usable as an insurer rating input, with demand patterns and a net-flow classifier for rebalancing.

Python · pandas · GeoPandas · risk analysis

Frequency-Report Scoring Rules

Inference & simulation

Belief elicitation recast as partial identification: characterized the identified set of beliefs behind each frequency report under three scoring rules and compared the sharpness of their bounds, validated by simulation.

Python · simulation · pytest

Economic Theories & Machine Learning

Model comparison

Benchmarked theory-based behavioral specifications against machine-learning models to measure how much of the predictable variation in choice data each theory actually captures.

Python · scikit-learn · model evaluation

RAG Search Engine

Retrieval system

Hybrid search combining BM25, embeddings, CLIP, and reciprocal-rank fusion with reranking and RAG answers, evaluated on precision, recall, and F1 against a golden set.

Python · embeddings · CLIP · evaluation

TEACHING

University lecturer and tutor in applied, behavioral, and organizational economics; supervised 60+ bachelor's and master's theses.

Seminar on Current Topics in Microeconomics

2025 – present

Undergraduate lecturer

Research Module on Applied Microeconomics & Management

2022 – 2025

Graduate lecturer

Empirical Evaluation of Management Practices

2019 – 2022

Graduate lecturer

Behavioral Management Science

2017 – 2020

Undergraduate lecturer

SKILLS

Programming Python · R · SQL · Stata · TypeScript

Python stack NumPy · pandas · SciPy · scikit-learn · statsmodels · PyTorch
· XGBoost / LightGBM · GeoPandas

Methods Statistical modeling · Econometrics · Machine learning
· Model evaluation & calibration · Forecasting · Experiment & A/B analysis
· Simulation · Fine-tuning · Retrieval / RAG

Tools Git · GitHub Actions · Docker · pytest · uv · JupyterLab · Linux · LaTeX

Languages English (C2) · German (C1/C2) · Danish (native)